

Spring 2021, MATH 8110 Real Analysis II, Midterm 2

PRINT NAME: _____ **ID#:** _____

- Show your work. Answers with no or incorrect steps will not receive partial credit.
 - The exam is self-explanatory. The instructor will not interpret the questions.
1. (4 points) Let the linear spaces $X = C^1([0, 1])$ and $Y = C([0, 1])$ be both endowed with norm $\|x\| := \max_{0 \leq t \leq 1} |x(t)|$. Define $T : X \rightarrow Y$ by $(Tx)(t) = x'(t)$ (x' the derivative of x) for all $t \in [0, 1]$ and $x \in X$. Prove that
- (a) T is a closed operator.
 - (b) T is not a continuous operator.

2. (4 points) Let $(X, \langle \cdot, \cdot \rangle)$ be a Hilbert space and M a linear subspace of X . Recall that $M^\perp := \{y \in X : \langle x, y \rangle = 0, \forall x \in M\}$. Prove that
- (a) $M^\perp$ is a closed linear subspace of X .
 - (b) $\bar{M} = M^{\perp\perp}$ where $M^{\perp\perp} := (M^\perp)^\perp$. [Hint: notice that $\bar{M}^\perp \subset M^\perp$.]

3. (4 points) Let X be a Banach space, E a nonempty open subset of X , and F a subset of first category in X . Show that $E \setminus F$ is dense in E . [Hint: It suffices to show $E \subset \overline{E \setminus F}$. To this end, consider to prove by contradiction while noticing that nonempty open sets are of second category.]

4. (4 points) Let X and Y be Banach spaces, $T_k \in L(X, Y)$ for all $k \in \mathbb{N}$. For any $x \in X$, the sequence $\{T_k x\}$ is convergent in Y . Show that there exists $T \in L(X, Y)$ such that $T_k x \rightarrow Tx$ for any $x \in X$ and $\|T\| \leq \liminf_{k \rightarrow \infty} \|T_k\| < \infty$. [Hint: use uniform boundedness theorem to show $\liminf_{k \rightarrow \infty} \|T_k\| < \infty$.]

5. (4 points) Let X be a Hilbert space and $T : X \rightarrow X$ be linear and self adjoint, i.e., $\langle Tx, y \rangle = \langle x, Ty \rangle$ for all $x, y \in X$. Show that T is continuous. [Hint: use closed graph theorem.]

6. (4 points bonus) Mark each of the following assertions True (T) or False (F).

- (a) ☐ The set $\{x \in \mathbb{R} : \sin(x) = 0\}$ is of first category in $\mathbb{R}$.
- (b) ☐ The set $\{(x, y) \in \mathbb{R}^2 : x^2 + y^2 = 1\}$ is of second category in $\mathbb{R}^2$.
- (c) ☐ Let X and Y be Banach spaces and $T : X \rightarrow Y$ be linear. Then T is bounded if and only if T is continuous at $0 \in X$.
- (d) ☐ Let X and Y be Banach spaces and $T \in L(X, Y)$. If T is surjective, then there exists $\delta > 0$ such that $U(0; \delta) \subset T(B(0; 1))$ where $B(0; r)$ and $U(0; r)$ denote the open balls centered at origin with radius r in X and Y respectively.

All problems above are shown on the page.

1. (4 points) Let the linear spaces $X = C^1([0, 1])$ and $Y = C([0, 1])$ be both endowed with norm $\|x\| := \max_{0 \leq t \leq 1} |x(t)|$. Define $T : X \rightarrow Y$ by $(Tx)(t) = x'(t)$ (x' the derivative of x) for all $t \in [0, 1]$ and $x \in X$. Prove that
 - (a) T is a closed operator.
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 - (a) $M^\perp$ is a closed linear subspace of X .
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4. (4 points) Let X and Y be Banach spaces, $T_k \in L(X, Y)$ for all $k \in \mathbb{N}$. For any $x \in X$, the sequence $\{T_k x\}$ is convergent in Y . Show that there exists $T \in L(X, Y)$ such that $T_k x \rightarrow Tx$ for any $x \in X$ and $\|T\| \leq \liminf_{k \rightarrow \infty} \|T_k\| < \infty$. [Hint: use uniform boundedness theorem to show $\liminf_{k \rightarrow \infty} \|T_k\| < \infty$.]
5. (4 points) Let X be a Hilbert space and $T : X \rightarrow X$ be self adjoint, i.e., $\langle Tx, y \rangle = \langle x, Ty \rangle$ for all $x, y \in X$. Show that T is continuous. [Hint: use closed graph theorem.]
6. (4 points bonus) Mark each of the following assertions True (T) or False (F).
 - (a) ☐ The set $\{x \in \mathbb{R} : \sin(x) = 0\}$ is of first category in $\mathbb{R}$.
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